



Primary Packaging Glass

Gerresheimer Tubular Glass Americas

Morganton Plant
114 Wamsutta Mill Road
Morganton, NC 28655 USA

James McFarland
Senior Regulatory Manager

Telephone 1.828.604.1429
E-Mail james.mcfarland@gerresheimer.com
Internet www.gerresheimer.com

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RE: Gerresheimer Tubular Glass Regulatory Summary

We confirm that our Type I flint and Amber glass containers comply with the current European Pharmacopoeia (EP), the United States Pharmacopoeia (USP) and the Japanese Pharmacopoeia (JP). Gerresheimer is providing this information in response to customer questionnaires and surveys. If you have additional compliance questions, please contact your Gerresheimer representative. This revision shall remain valid unless a change is made to a product or process impacting the statements herein.

Tubular glass composition:

Gerresheimer Glass has the ability to source tubing from suppliers around the globe. The composition of the tube is specific to each source and information for chemical and physical properties are available on request for your specific product.

Elemental Impurities ICH Q3D:

We confirm that our products are produced according to the Guideline for Elemental Impurities, dated October 3, 2020.

ICH Q3D guidelines apply to applicants of final drug product but recognize direct and/or indirect sources of metals from excipients, catalysts, from components of container closure systems and drug manufacturing equipment.

Gerresheimer Tubular Glass Americas certifies that it does not intentionally add arsenic, cadmium, lead, mercury, and/or hexavalent chrome to any glass formulation. Additionally, the sum of any incidental level of these substances does not exceed 100 ppm within any given container.

TSE / BSE

This is to certify that the products produced by Gerresheimer Tubular Glass Americas are produced entirely free of animal derived materials including bovine products. No animal derived components are used in the manufacturing or handling processes of this product. As such, this material can be declared free of Bovine Spongiform Encephalopathy (BSE) and Transmissible Spongiform Encephalopathy (TSE).

EU REACH Regulation and SVHC in particular to Glass Articles

Gerresheimer Tubular Glass Americas considers the protection of the environment and natural resources as one of its most important business tasks and devotes significant efforts and resources to the development of clean products and processes, and thereby concurs with the aim of the European Union's Registration, Evaluation, Authorization and Restriction of Chemicals (REACH) Regulation (1907/2006/EC) including the 240 Substances of Very High Concern as of January 23, 2024 as well as chemicals listed in REACH Annex XVII.

Glass products produced by Gerresheimer Tubular Glass Americas, do not contain substances that are intended to be released under normal and foreseeable circumstances and are therefore not concerned by Article 7.1 of the

Reach Regulation.

Concerning Article 7.2 of the Reach Regulation, which requires notification of substances in products, if such substances are of high concern, exceed 0.1% (w/w) and exceed 1 ton/year per importer or producer, Article 33.1 of the REACH Regulation requires a supplier to inform customers if the article contains a substance of very high concern (SVHC – substance potentially to be included in Annex XVII). The following should be noted:

Glass is treated as a UVCB substance¹ under REACH and is in principle exempted from registration under the REACH Regulation (see Annex II of Commission Regulation 987/2008 for Annex V exemption 11).

Furthermore, glass is not classified as a hazardous substance and it is unlikely ever to be listed on the list of substances of high concern.

We confirm that Gerresheimer Tubular Glass Americas, Type 1 glass complies with the above regulations and the dangerous substances listed are not part of our glass finished product.

RoHS

European Directive 2015/863 of 22 July 2019 on the restriction of the use of certain hazardous substances in electrical and electronic equipment (RoHS 3).

Under the RoHS Directive the use of ten substances, Cadmium: < (0.01 %), Lead: < (0.1 %) Mercury: < (0.1 %), Hexavalent chromium: < (0.1 %), Polybrominated biphenyls (PBB): < (0.1 %) Polybrominated diphenyl ethers (PBDE): < (0.1 %), Bis(2-ethylhexyl) phthalate (DEHP): < (0.1 %) Benzyl butyl phthalate (BBP): < 1000 ppm (0.1%) Dibutyl phthalate (DBP): < 1000 ppm (0.1%) Diisobutyl phthalate (DIBP): < 1000 ppm (0.1%), in electrical and electronic equipment put onto the European market after 1 July 2006, is subject to the stated maximum concentration values.

We would like to inform you that our Gerresheimer Tubular Glass Americas products do not contain any of the above cited substances above the limit values. In addition, Gerresheimer Tubular Glass Americas, certifies that it does not intentionally add cadmium, lead, mercury, hexavalent chrome, PBDE's and or PBB's to any glass formulation.

Melamine Declaration

Gerresheimer Tubular Glass Americas manufactures tubular glass containers. Melamine or Melamine containing materials are not batched into any Gerresheimer Glass tube formulation. Additionally, glass is melted in excess of 2000° F, containers are annealed in excess of 1000° F and the melting point of Melamine is approximately 650° F. Tubular glass containers manufactured by Gerresheimer Tubular Glass Americas are not classified as at-risk components for Melamine. Based on the above statements, Melamine testing is not required. Glass products manufactured by Gerresheimer Tubular Glass Americas can be certified to be "melamine free".

Food Regulations

Please be informed that none of the following food or food additives/preservatives are present in Gerresheimer Tubular Glass Americas, Type 1 containers:

Alcohol, Ethyl alcohol, Antioxidant food substances, Corn Starch, Egg/Protein, Fish/Shellfish, Genetically Modified Organisms (GMO), Gluten, Halal / Non-Halal Food Products or ingredients, Lactose, Metabisulphite, Nuts, including peanuts, Preservatives, Red Dye, Sugar, Soy, Nitrosamines.

Additionally, as per the most current version of ISO 9001 (Primary packaging materials) requirements, all food products are restricted from the glass production lines where direct contact with glass products is possible.

PROPOSITION 65 Safe Drinking Water and Toxic Enforcement Act of 1986

California's Proposition 65 is a list of chemicals known to the state to cause cancer or reproductive toxicity. List as dated April 11, 2023. Gerresheimer Tubular Glass Americas has evaluated its Type 1 glass regarding items on the California Proposition 65 List of Chemicals. All ingredients batched into the specific glass formulations become fused into the specific glass matrix and therefore, are not airborne or "free."

Gerresheimer Tubular Glass Americas has performed extractable testing to evaluate Gerresheimer glasses regarding Heavy Metals and performs the USP / EP required testing for arsenic extraction. Gerresheimer Glass

does not intentionally add any lead, cadmium, chrome, mercury, arsenic, or antimony in its glass formulation. Our results for USP and EP arsenic extraction are below detectable limits and recorded as <0.001 ppm.

Gerresheimer Tubular Glass Americas has determined that the glass (fused matrix) containers do not have any significant exposures and no warning is required for our product under the guidelines presented for the California Proposition 65.

CONEG “Toxics in Packaging Clearinghouse” (TPCH) model legislation on Packaging and Packaging Waste

The Model Toxics in Packaging Legislation was jointly developed by representatives for the Governors of the nine (9) Northeast States, major national product producers, major national retailers and national and regional environmental organizations in 1989 working under the auspices of the Coalition of Northeast Governors (CONEG) Policy Research Center. The intent was for individual Northeast states to adopt the Model Legislation, and thereby, create consistent requirements for the reduction/elimination of four (4) specific toxics (specifically, mercury, lead, cadmium, and hexavalent chromium) in packaging sold or distributed in the Northeast. The concentration level of heavy metals (Cd (Cadmium), Pb (Lead), Hg (Mercury), and CR VI (Hexavalent Chrome)) in Gerresheimer Glass Type 1 containers is below 100ppm by weight (0.01%) as required in above mentioned legislation. Heavy metals (Cd, Pb, Hg, Cr VI) are not intentionally added to the batch formula of the glass.

EU Packaging and Packaging Waste Directive

European Union Directive 94/62/EC, “Packaging and Packaging Waste Directive”, contains a specific limit (<100ppm) for the total content of four heavy metals (cadmium, hexavalent chromium, mercury, and lead) by weight that is allowed to be present in packaging and packaging materials. Furthermore, the directive requires assessing all packaging systems regarding the so-called “essential requirements” and the applicable harmonized European Standards.

We confirm that the packaging used for your Type 1 containers, produced by Gerresheimer Tubular Glass Americas complies with the limitation on heavy metals and the essential requirements.

Plastic and Rubber substances

Please be informed that none of the following substances are present in Gerresheimer Tubular Glass Americas, Type 1 containers:

Antioxidant Substances, Benzophenone, Butylated Hydroxytoluene (BHT), Bisphenol A (BPA), Latex Mercaptobenzothiazole (2-MBT, 2-MCBT), Natural rubber latex, Natural rubber, Nitrosamines, Phthalates (DINP, DIDP, DEHP, BBP, DBP), Polystyrene, Silicone, PVC.

Exception are vials that are purposely treated with silicone for interior lubrication of the tubular glass container.

Conflict Minerals (Dodd-Frank Act)

In 2010, Congress passed the Dodd-Frank Act which directs the Commission to issue rules requiring certain companies to disclose their use of conflict minerals if those minerals are “necessary to the functionality or production of a product” manufactured by those companies. Under the Act, those minerals are gold, wolframite, cassiterite, columbite-tantalite (coltan), tin, tungsten, or tantalum. Congress enacted Section 1502 of the Act because of concerns that the exploitation and trade of conflict minerals by armed groups is helping to finance conflict in the DRC region (Democratic Republic of the Congo) and is contributing to an emergency humanitarian crisis.

Gerresheimer Tubular Glass Americas confirms that all components are certified free of Conflict Minerals and therefore are not intentionally added during any part of the process.

Nitrosamines Statement of Compliance

Gerresheimer Tubular Glass Americas has evaluated its sources, materials, and processes regarding nitrosamines contamination risk. Type 1 borosilicate glass is melted using a batch comprising inorganic minerals at temperatures exceeding 2000 degrees Fahrenheit. We confirm that based upon third party verification, Type I Borosilicate Glass is nitrosamine free. No nitrosamines are intentionally added or used in any raw material or processing step during the production of glass containers by Gerresheimer Tubular Glass Americas.

Final processing of glass containers (for pharma applications)

We confirm that our final products are produced, controlled, and packed in an environmentally controlled area per ISO 9001 and ISO 15378 standard requirements.

Storage conditions and Shelf Life, Humidity and Temperature

Type I Borosilicate glass is used to manufacture tubular containers produced by Gerresheimer Tubular Glass Americas. The glass is formulated to be very resistant to chemical attack. Gerresheimer glass determines the chemical resistance of this glass as part of its routine glass composition and property control program. Due to the nature of Borosilicate glass and the glass's high resistance to chemical deterioration during storage, shelf life is "indeterminate".

There are no specific requirements for storage, other than general requirements. There is no shelf life specified, and there is no established re-analysis time frame.

CMR, Endocrine Disruptors, PBT, vPvB, and Nanomaterials

Type I Borosilicate glass is used to manufacture tubular containers produced by Gerresheimer Tubular Glass Americas do not contain CMR (carcinogens, mutagens or reprotoxic substances that are toxic for reproduction, as listed in Annex VI of Regulation (EC) 1272/2008 on Classification, labeling and packaging of substances and mixtures (CLP), or nanomaterials as defined by the EU "Commission Recommendation 2011/696/EU".

Gerresheimer Tubular Glass containers do not contain Endocrine disruptors (i.e., BPA, Dioxins, Perchlorate, Perfluoroalkyl and Polyfluoroalkyl Substances (PFAS), Phthalates, Phytoestrogens, Polybrominated diphenyl ethers (PBDE), Polychlorinated biphenyls (PCB), Triclosan), PBT (Persistent Bio accumulative and Toxic) or vPvB (very active substances in quantity equal or above 0.1% (w/w) as listed in REACH or SVHC (See REACH statement).

Debarment

Gerresheimer Tubular Glass herewith confirms that, to the best of its knowledge, it does not use, in any capacity, the services of any person debarred under sections 306 (a) or (b) of the Federal Food, Drug, and Cosmetic Act (21 U.S.C. 335(a) or (b)) with its manufacture of Tubular Glass containers.

GRAS – Generally recognized as Safe

Gerresheimer Tubular Glass containers meet the Type I pharmaceutical primary packaging container criteria for Type I glass according to the European Pharmacopoeia Chapter 3.2.1 and United States Pharmacopoeia Chapter <660>. The metals Cd (cadmium), Pb (Lead), Hg (mercury), Cr VI (hexavalent Chromium) As (arsenic) or Sb (antimony) are not added intentionally.

According to the US FDA (Federal Register / Vol. 48, No. 27 / Feb 08, 1983 / page 5718, Rules and Regulations) glass is considered as GRAS, Generally Recognized as Safe.

Manufacturing exclusivity / Risk Mitigation

Antibiotics- We confirm that our process does not involve any antibiotics.

Radiation – We confirm that we do not Radiate any of our products nor do our suppliers radiate our materials supplied to us.

Environmental contaminates – There are no polychlorinated biphenyl or polybrominated biphenyl compounds produced as by products in the manufacturing of our glass.

Doping – Our glass products do not contain any substances that would be considered doping nor is the glass itself considered doping.

Water Source

Non-Ready to fill vials – Water is only used to control temperatures in the manufacturing process and does not come into contact with the product. No testing is required.

Ready to Fill vials – Water for injection is used to wash the vials as part of the manufacturing process. Testing of the water is conducted on a regular basis.

POP compliance

The products that Gerresheimer Glass manufactures do not contain any of the organic pollutants listed by the

Persistent Organic Pollutant Regulation (POPs) covered by regulation (EU) 2019/1021. This regulation contains a list of organic pollutants that could impact the environment, or accumulate in living organisms and pose a risk to our health and safety.

General remarks

The above statements are based on information available to Gerresheimer at this point in time. Although Gerresheimer believes the statements are accurate, it does not warrant their accuracy, and Gerresheimer will not be liable for any reliance on the statements. Gerresheimer is providing these statements in good faith, but is not

warranting their completeness or accuracy, and it does not accept liability for damages resulting from the use or reliance upon these statements.

Should you have any further questions, please do not hesitate to contact us.

Yours sincerely,

A handwritten signature in black ink, appearing to read "James McDaniel". The signature is fluid and cursive, with the first name "James" being more legible than the last name "McDaniel".

Senior Regulatory Manager
Gerresheimer Tubular Glass Americas